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(54) **AUTOMATIC DRAWER-OPENING DEVICE**

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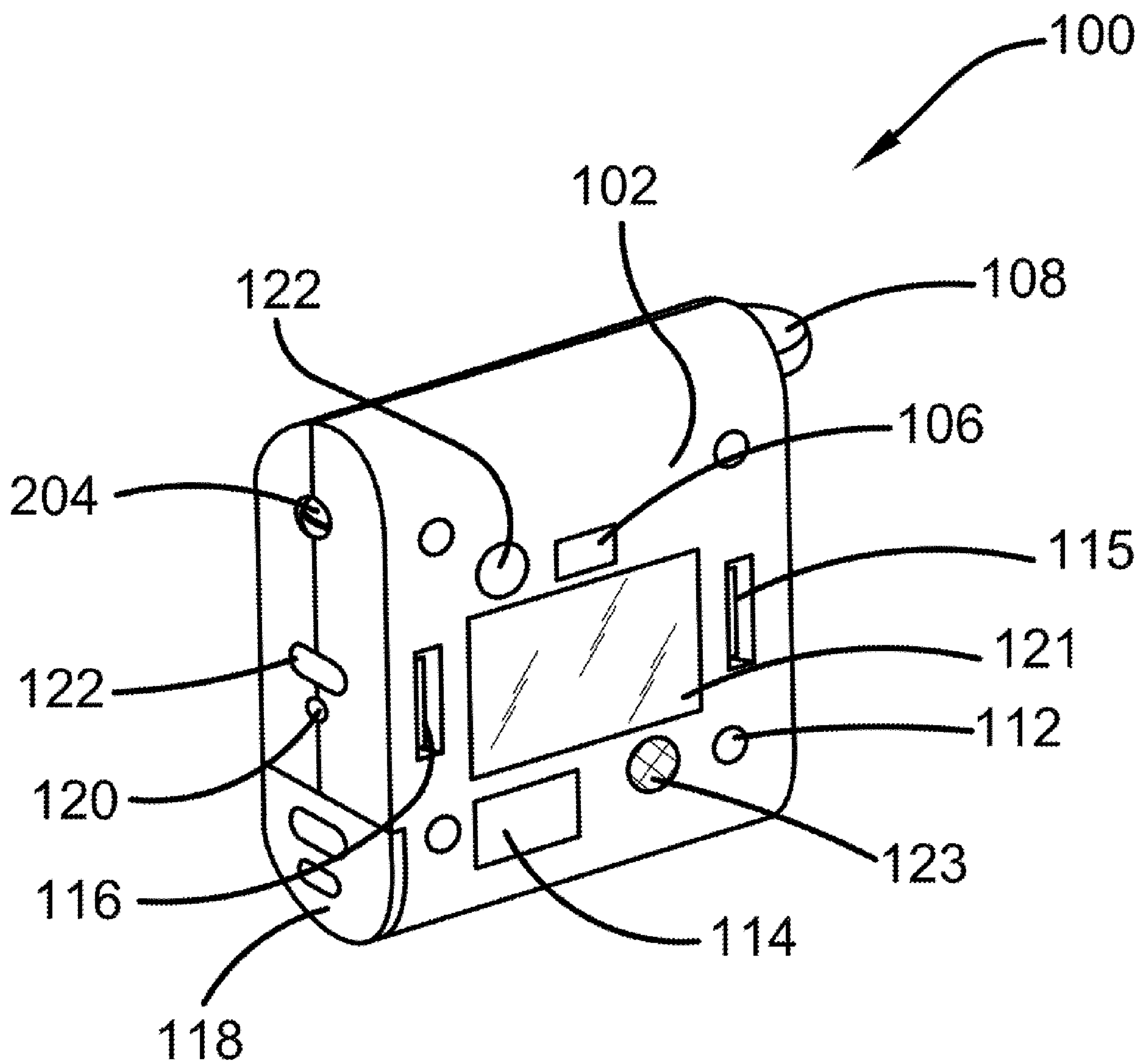
(57) **ABSTRACT**

An automatic drawer-opening device is disclosed, which is designed to automate the operation of the opening and closing of drawers. The automatic drawer-opening device comprises a housing component that is configured to house a processor and a battery-operated motor that is engaged via a hidden proximity sensor or wireless controls. Specifically, the motor can be installed within the wheel and is preferably a DC motor with a spring-loaded wheel. The spring-loaded wheel has a rubber coating for smooth operation and provides for adjustable tension for user-specific customization. Further, the motor can sync with smart features via Wi-Fi, Bluetooth, or virtual assistants.



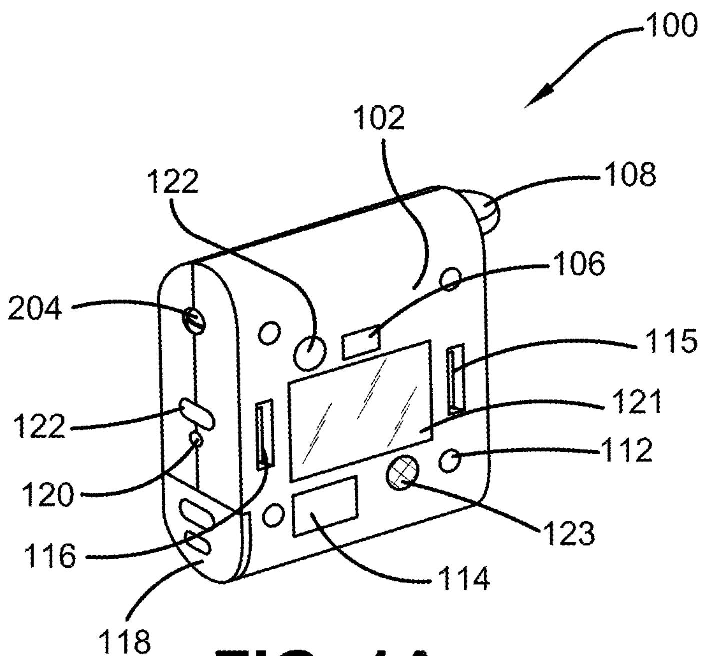

100
102
122
108
106
204
115
121
122
112
120
123
116
114
118

FIG. 1A

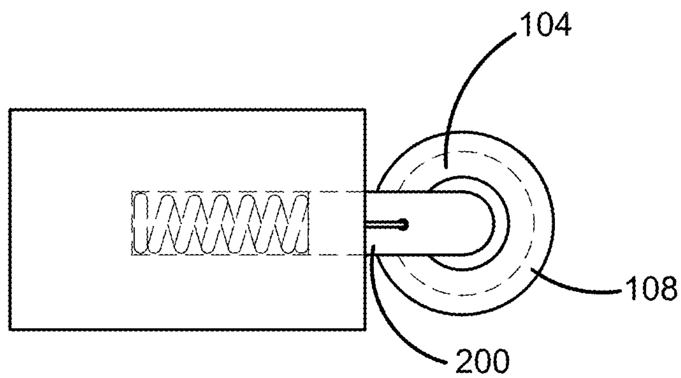

104
108
200

FIG. 1B

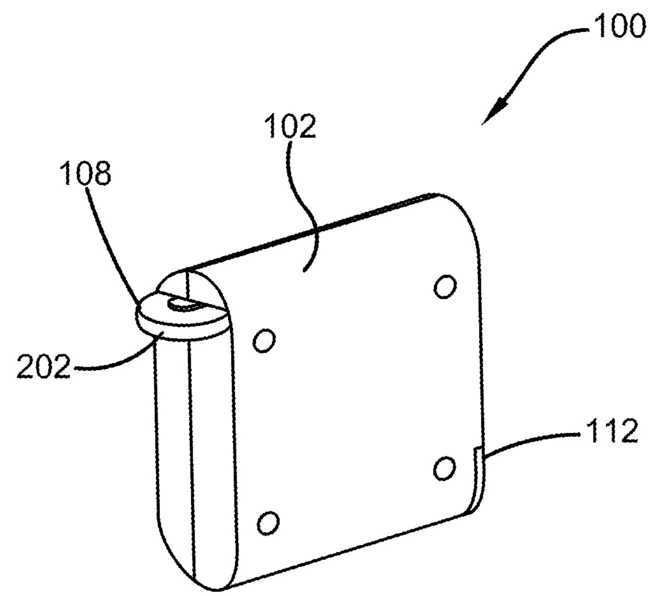


FIG. 2A

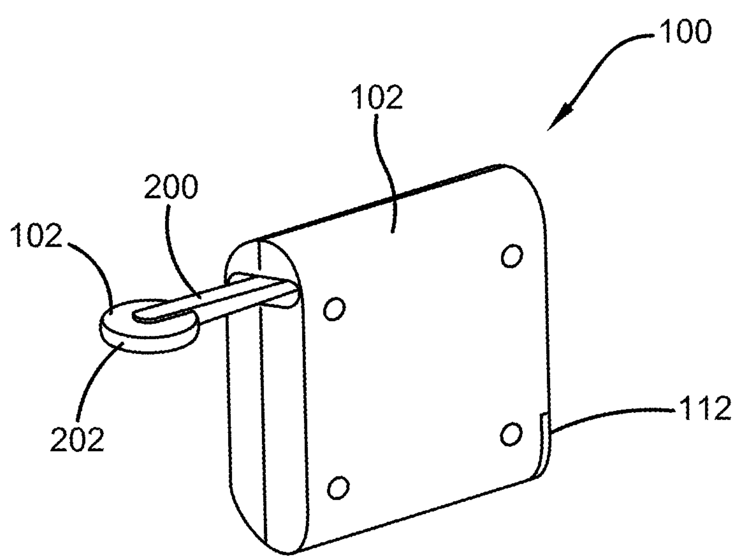


FIG. 2B

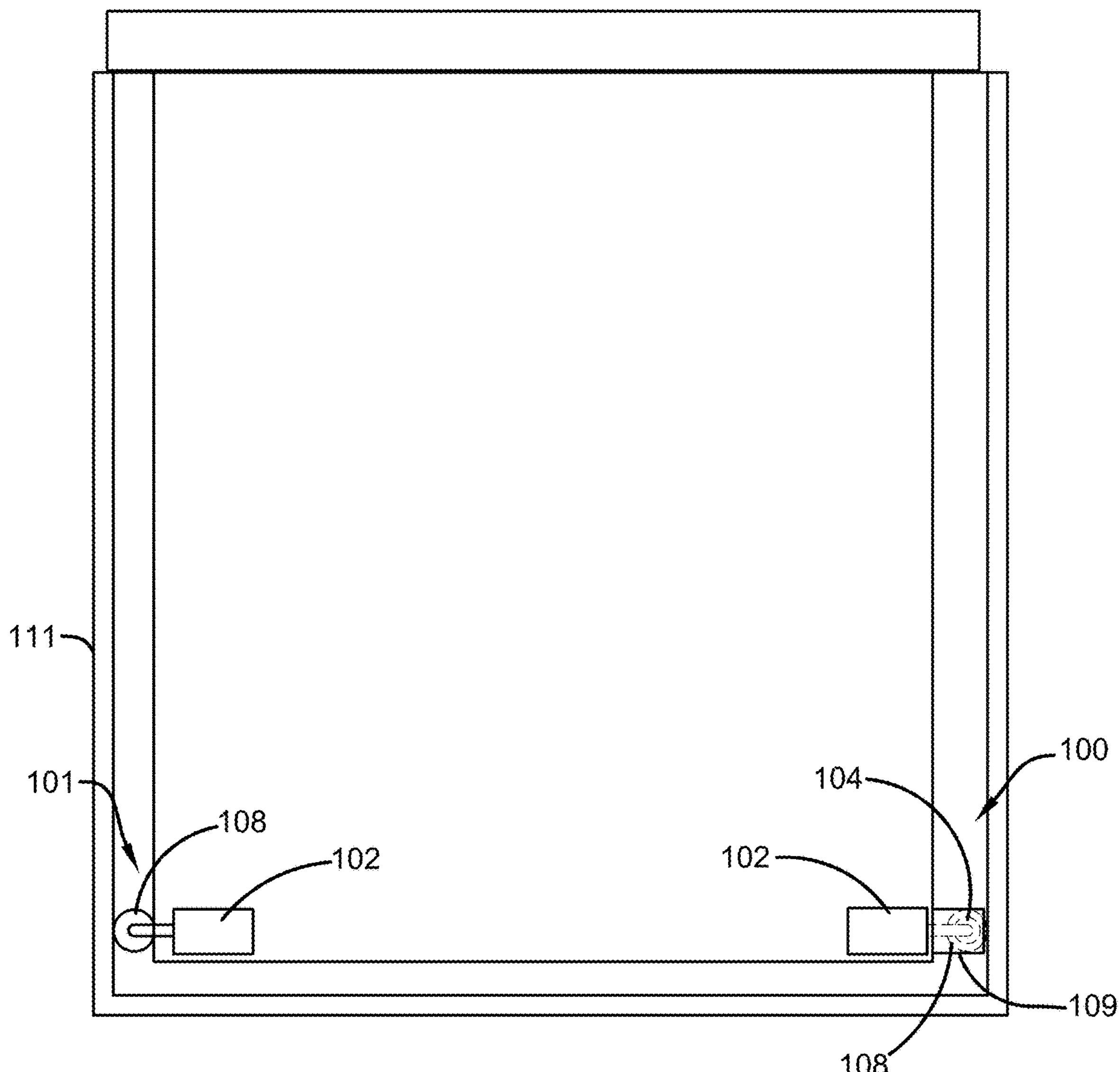


FIG. 3

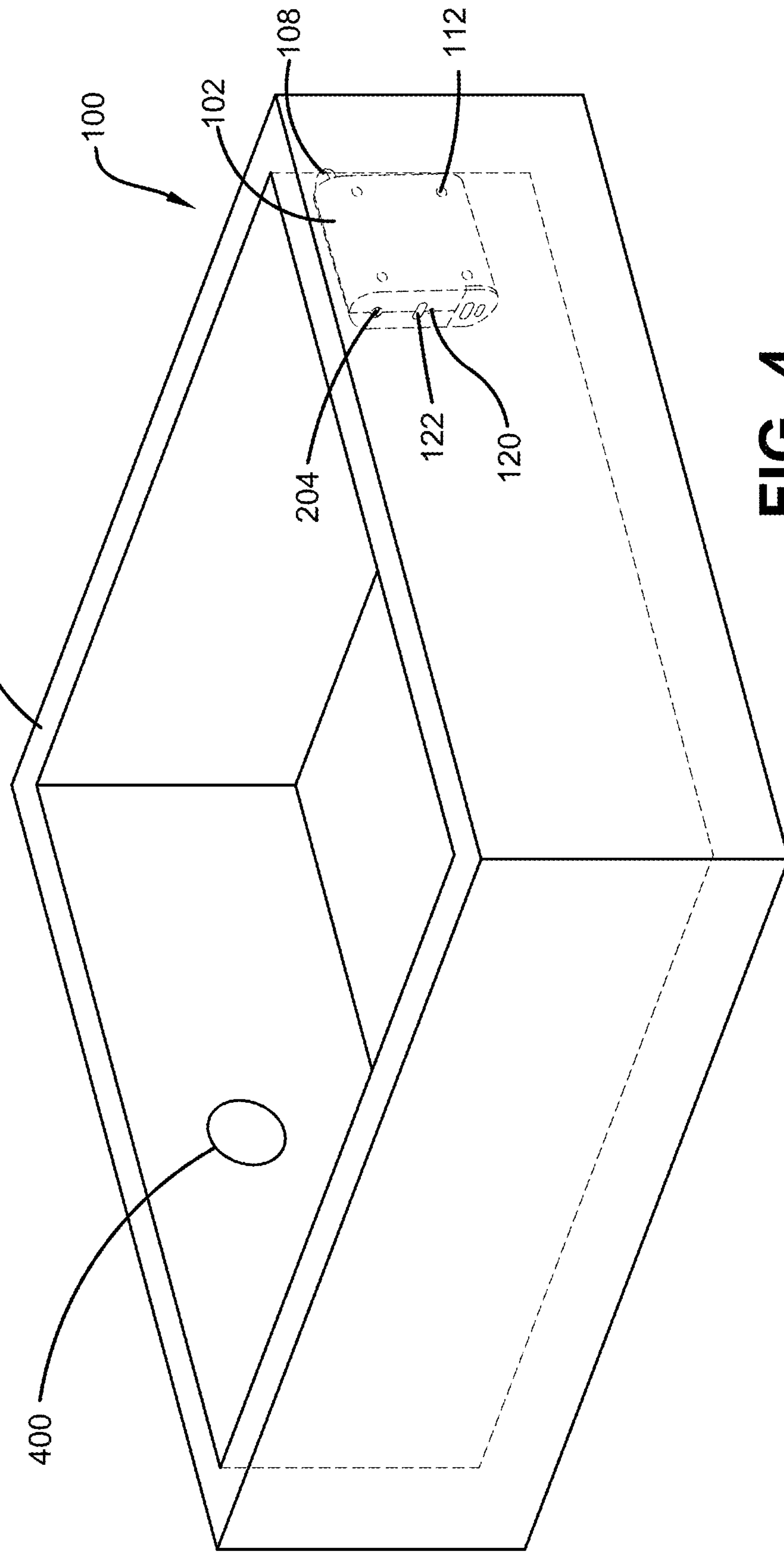


FIG. 4

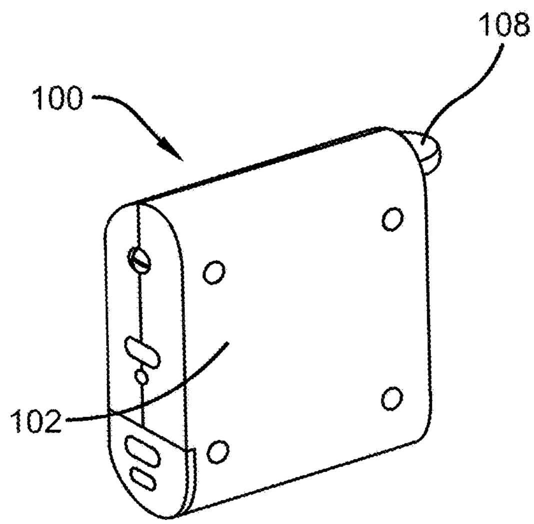


FIG. 5A

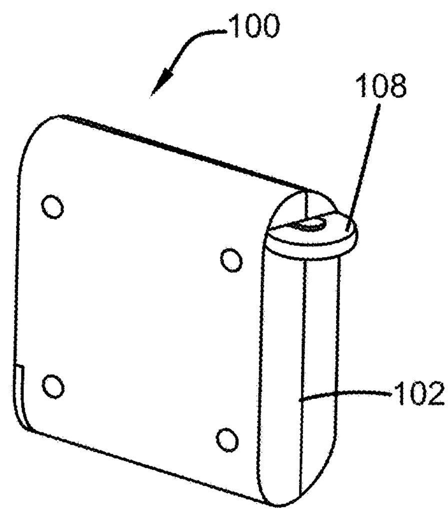


FIG. 5B

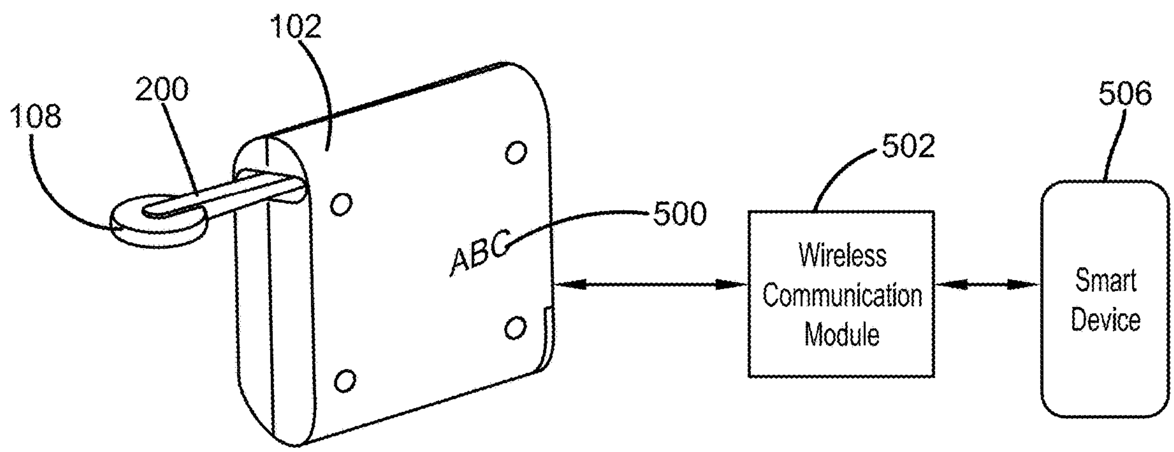


FIG. 5C

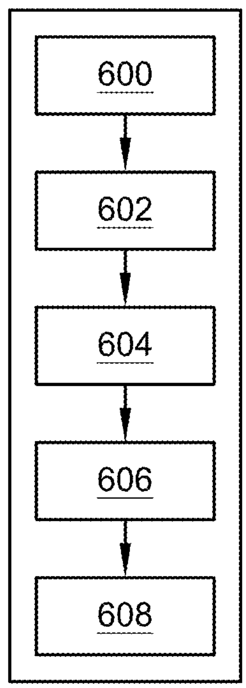


FIG. 6

AUTOMATIC DRAWER-OPENING DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to, and the benefit of, U.S. Provisional Application No. 63/556,928, which was filed on Feb. 23, 2024, and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates generally to the field of automatic drawer-opening devices. More specifically, the present invention relates to a device for opening drawers, that allows a drawer to be opened in a hands-free manner. Accordingly, the present disclosure makes specific reference thereto. Nonetheless, it is to be appreciated that aspects of the present invention are also equally applicable to other like applications, devices and methods of manufacture.

BACKGROUND

[0003] By way of background, this invention relates to improvements in automatic drawer-opening devices. Generally, in many circumstances, an individual may struggle to or wish to avoid opening a drawer with their hand. This could be due to a number of factors, such as dirty hands or having no free hand available.

[0004] Further, some users may have grip issues and thus, may have trouble opening drawers. For example, users with disabilities and elderly users would benefit from an automatic drawer opener. Additionally, an automatic drawer opener eliminates the need for drawer pulls and creates a clean aesthetic for the drawers.

[0005] Accordingly, there is a demand for an improved automatic drawer-opening device that allows a drawer to be opened in a hands-free manner. More particularly, there is a demand for an automatic drawer-opening device that can sync with smart features via Wi-Fi, Bluetooth or virtual assistants.

[0006] Therefore, there exists a long felt need in the art for an automatic drawer-opening device that allows a drawer to be opened in a hands-free manner. There is also a long felt need in the art for an automatic drawer-opening device that can be installed in the back of a drawer. Further, there is a long felt need in the art for an automatic drawer-opening device that includes a rubber coating on the spring-loaded wheel for smooth operation. Moreover, there is a long felt need in the art for a device that includes adjustable tension for user-specific customization. Further, there is a long felt need in the art for an automatic drawer-opening device that can sync with smart features via Wi-Fi, Bluetooth, or virtual assistants. Finally, there is a long felt need in the art for an automatic drawer-opening device that can be installed aftermarket for home DIY installations.

[0007] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises an automatic drawer-opening device. The device is designed to automate the operation of opening and closing of drawers. The automatic drawer-opening device comprises a housing component that is configured to house a processor and a battery-operated motor that is engaged via a hidden proximity sensor or wireless controls. Specifically, the motor can be housed within the wheel and is preferably a DC motor with a spring-loaded wheel. The spring-loaded wheel has a rubber coating for smooth operation and provides for adjustable tension for user-specific customization. Further, the motor can sync with smart features via Wi-Fi, Bluetooth, or virtual assistants.

[0008] In this manner, the automatic drawer-opening device of the present invention accomplishes all of the forgoing objectives and provides users with a device that allows a drawer to be opened in a hands-free manner. The device has a battery-operated motor engaged through a sensor or wireless controls.

SUMMARY OF THE INVENTION

[0009] The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to present some general concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0010] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises an automatic drawer-opening device. The device is designed to automate the operation of opening and closing of drawers. The automatic drawer-opening device comprises a housing component with a processor and a battery-operated motor, which is installed in the spring-loaded wheel. The motor acts to rotate the spring-loaded wheel for opening and/or closing the drawer hands-free.

[0011] In one embodiment, the automatic drawer-opening device allows a drawer to be opened in a hands-free manner. As users may struggle to open a drawer due to having their hands full or having dirty hands, other users may may have grip issues, such as people with disabilities and the elderly. The automatic drawer-opening device eliminates the need for drawer pulls and automatically opens and closes drawers, as needed.

[0012] In one embodiment, the automatic drawer-opening device is utilized to mainly open and/or close a kitchen drawer automatically via a motor. The device fits most common drawer styles, as is known in the art. Furthermore, the automatic drawer-opening device can be utilized in other similar applications, such as dresser drawers, bathroom drawers, etc.

[0013] In one embodiment, the automatic drawer-opening device comprises a housing component which is configured in a size and shape to fit in the back of a kitchen drawer, etc., during use. Further, the housing component comprises a plurality of mounting holes for allowing a user to insert a screw and mount the device in position in the back of a kitchen drawer. Generally, the housing component is typically a rectangular configuration, but can be any suitable size and shape, such as square, oval, circular, etc. Further, the housing component can be any suitable shape and size as is known in the art, as long as the housing component is sized and shaped to house the components and circuitry needed for utilizing the device to automatically open and close the drawers the device is secured to.

[0014] In one embodiment, the housing component is configured to house a processor and a motor. Specifically, the housing component is configured to house the processor and motor and any other circuitry as is known in the art for engaging the motor to automatically open and close a kitchen drawer.

[0015] In one embodiment, the motor is a DC motor but can be any suitable motor as is known in the art. In use, the motor can be installed within the spring-loaded wheel to rotate it or other suitable device for contacting the drawer to automatically open and/or close it.

[0016] In one embodiment, the spring-loaded wheel is controlled by the motor to rotate it, such that the wheel can spin. Specifically, the wheel is secured to a spring-loaded arm, which is tensioned to keep the arm pushed against the cabinet, allowing the wheel to contact the drawer. When activated by the motor, the wheel spins to ride the cabinet enclosure to open and close the drawer hands-free. Then, once open, the motor is engaged again to spin the spring-loaded wheel in the opposite direction, to close the drawer hands-free. In one embodiment, the spring-loaded wheel has a rubber coating for smooth operation and to protect the drawer it comes into contact with. In one embodiment, the spring-loaded wheel can have adjustable tension, depending on the weight of the drawer being opened and/or closed. Thus, the device can accommodate different drawer weights. Specifically, the housing component comprises a tension adjustment screw, which allows a user to turn the screw right or left to provide for adjustable tension and user-specific customization.

[0017] In one embodiment, the motor is battery-operated and wireless. In this embodiment, the battery can be replaceable or rechargeable. If rechargeable, the battery can be recharged via a USB port positioned on the housing component. In one embodiment, the housing component comprises a battery door, to allow a user access to the batteries, as needed. The housing component also comprises multiple function buttons for controlling the device, as needed. For example, the housing component comprises a multicolor LED to let a user know if the power to the device is turned on or not. For example, the LED light may show green when on, red when off, and yellow when the device needs charged or any other suitable colors as is known in the art. The housing component can also comprise a multifunction button which can be a power button or can control any other suitable functions on the device, as needed.

[0018] In one embodiment, the motor is engaged via a hidden proximity sensor. Typically, the hidden proximity sensor is positioned inside the drawer face or on an outside of the drawer to be activated by a user who comes in close proximity to the drawer to be opened. The proximity sensor can be any suitable sensor as is known in the art, and is wired or wireless. The proximity sensor is in communication with the motor and processor, such that once the sensor is engaged, the sensor communicates with the motor to rotate the spring-loaded wheel.

[0019] In one embodiment, the motor is controlled by wireless controls. Specifically, the motor can sync with smart features via Wi-Fi, Bluetooth, or virtual assistants. In this embodiment, the device further includes a wireless communications module and additional sensors which would allow the device to pair with a mobile application on a smart device. Once paired, a user could control the automatic drawer-opening device via the mobile application.

[0020] In one embodiment, the automatic drawer-opening device can be installed at the point of manufacture or installation of the drawers. In another embodiment, the device can be installed aftermarket, after the drawers have been installed. Further, the device is great for DIY installations and is easy-to-install, as needed. Thus, the automatic drawer-opening device is great for users with grip issues, people with disabilities, or elderly users. Further, the device eliminates the need for drawer pulls and displays a cleaner aesthetic, offering a clean, sleek appearance for drawers.

[0021] In one embodiment, multiple devices can be linked together to operate in sync if the drawer is large and heavy. One device would be installed on the right side of the drawer and a second device on the left side of the drawer. With either the first or the second device, there may be a non-electrical spring-loaded wheel which mounts opposite the electrical spring-loaded wheel to equalize pressing pressure of the spring-loaded wheels and to ensure the devices do not bind and operate smoothly.

[0022] In yet another embodiment, the automatic drawer-opening device comprises a plurality of indicia.

[0023] In yet another embodiment, a method of opening a drawer hands-free is disclosed. The method includes the steps of providing an automatic drawer-opening device comprising a housing component with a motor and spring-loaded wheel. The method also comprises installing the housing component in a back of a drawer. Further, the method comprises engaging the motor to rotate the spring-loaded wheel via a proximity sensor. The method also comprises syncing the motor with smart features via Wi-Fi, Bluetooth, or virtual assistants. Finally, the method comprises adjusting tension of the spring-loaded wheel to accommodate different drawer weights.

[0024] Numerous benefits and advantages of this invention will become apparent to those skilled in the art to which it pertains, upon reading and understanding the following detailed specification.

[0025] To the accomplishment of the foregoing and related ends, certain illustrative aspects of the disclosed innovation are described herein in connection with the following description and the annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and are intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:

[0027] FIGS. **1**A and **1**B illustrate a perspective view of one embodiment of the automatic drawer-opening device of the present invention in accordance with the disclosed architecture;

[0028] FIGS. **2**A and **2**B illustrate a perspective view of one embodiment of the automatic drawer-opening device of the present invention showing the roller wheel contacting the drawer in accordance with the disclosed architecture;

[0029] FIG. **3** illustrates a perspective view of one embodiment of the automatic drawer-opening device of the present invention showing multiple devices in one drawer in accordance with the disclosed architecture;

[0030] FIG. **4** illustrates a perspective view of one embodiment of the automatic drawer-opening device of the present invention showing the battery-powered DC motor in accordance with the disclosed architecture;

[0031] FIGS. **5**A, **5**B, and **5**C illustrate a perspective view of one embodiment of the automatic drawer-opening device of the present invention showing the device and its components in accordance with the disclosed architecture; and

[0032] FIG. **6** illustrates a flowchart showing the method of opening a drawer hands-free in accordance with the disclosed architecture.

DETAILED DESCRIPTION OF THE PRESENT INVENTION

[0033] The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They are not intended as an exhaustive description of the invention and do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

[0034] As noted above, there is a long felt need in the art for an automatic drawer-opening device that allows a drawer to be opened in a hands-free manner. There is also a long felt need in the art for an automatic drawer-opening device that can be installed in the back of a drawer. Further, there is a long felt need in the art for an automatic drawer-opening device that includes a rubber coating on the spring-loaded wheel for smooth operation. Moreover, there is a long felt need in the art for a device that includes adjustable tension for user-specific customization. Further, there is a long felt need in the art for an automatic drawer-opening device that can sync with smart features via Wi-Fi, Bluetooth, or virtual assistants. Finally, there is a long felt need in the art for an automatic drawer-opening device that can be installed aftermarket for home DIY installations.

[0035] The present invention, in one exemplary embodiment, is a novel automatic drawer-opening device. The device is designed to automate the operation of opening and closing of drawers. The automatic drawer-opening device comprises a housing component that is configured to house a processor and a battery-operated motor that is engaged via a hidden proximity sensor or wireless controls. Specifically, the motor can be installed in the wheel and is preferably a DC motor with a spring-loaded wheel. The spring-loaded wheel has a rubber coating for smooth operation and provides for adjustable tension for user-specific customization. Further, the motor can sync with smart features via Wi-Fi, Bluetooth, or virtual assistants. The present invention also includes a novel method of opening a drawer hands-free. The method includes the steps of providing an automatic drawer-opening device comprising a housing component with a motor and spring-loaded wheel. The method also comprises installing the housing component in a back of a drawer. Further, the method comprises engaging the motor to rotate the spring-loaded wheel via a proximity sensor. The method also comprises syncing the motor with smart features via Wi-Fi, Bluetooth, or virtual assistants. Finally, the method comprises adjusting tension of the spring-loaded wheel to accommodate different drawer weights.

[0036] Referring initially to the drawings, FIG. **1** illustrates a perspective view of one embodiment of the automatic drawer-opening device **100** of the present invention. In the present embodiment, the automatic drawer-opening device **100** is an improved automatic drawer-opening device **100** that allows a drawer to be opened in a hands-free manner. Specifically, the automatic drawer-opening device **100** comprises a housing component **102** that houses the battery-operated motor **104** and processor **106**, to control a spring-loaded wheel **108**. The motor **104** rotates a spring-loaded wheel **108** for opening and/or closing the drawer **110** hands-free.

[0037] Generally, the automatic drawer-opening device **100** allows a drawer **110** to be opened in a hands-free manner. As users may struggle to open a drawer **110** due to having their hands full or having dirty hands, other users may may have grip issues, such as people with disabilities and the elderly. The automatic drawer-opening device **100** eliminates the need for drawer pulls and automatically opens and closes drawers **110**, as needed.

[0038] Further, the automatic drawer-opening device **100** is utilized to mainly open and/or close a kitchen drawer **110** automatically via a motor **104**. The device **100** fits most common drawer styles, as is known in the art. Furthermore, the automatic drawer-opening device **100** can be utilized in other similar applications, such as dresser drawers, bathroom drawers, etc.

[0039] Additionally, the automatic drawer-opening device **100** comprises a housing component **102** which is configured in a size and shape to fit in the back of a kitchen drawer **110**, etc., during use. Further, the housing component **102** comprises a plurality of mounting holes **112** for allowing a user to insert a screw and mount the device **100** in position in the back of a kitchen drawer **110**. Generally, the housing component **102** is typically a rectangular configuration, but can be any suitable size and shape, such as square, oval, circular, etc. Further, the housing component **102** can be any suitable shape and size as is known in the art, as long as the housing component **102** is sized and shaped to house the components and circuitry needed for utilizing the device **100** to automatically open and close the drawers **110** the device **100** is secured to.

[0040] Typically, the motor **104** is battery-operated and wireless. In this embodiment, the battery **114** can be replaceable or rechargeable. If rechargeable, the battery **114** can be recharged via a USB port **116** positioned on the housing component **102**. Further, an external battery pack can be connected to the USB port **116** for providing extended run time of the device **100**. In one embodiment, the housing component **102** comprises a second port **115**, similar to a 3.5 mm headphone jack for plugging in sensors, i.e., a proximity sensor or a touch sensor. In one embodiment, the housing component **102** comprises a battery door **118**, to allow a user access to the batteries **114**, as needed. The housing component **102** also comprises multiple function buttons for controlling the device **100**, as needed. For example, the housing component **102** comprises a multicolor LED **120** to let a user know if the power to the device **100** is turned on or not. For example, the LED light **120** may show green when on, red when off, and yellow when the device **100** needs charged or any other suitable colors as is known in the art. The housing component **102** can also comprise a multifunction button

122 which can be a power button or can control any other suitable functions on the device 100, as needed. The multifunction button 122 allows a short press to turn on and off, and a long press to enter pairing mode. The housing component 102 can also comprise a small LCD screen 121 to display additional information, as needed. The housing component 102 also comprises a buzzer 123 to sound when the drawer 110 is left open after a certain amount of time or if there is an error with the device 100. Sound can also play from the buzzer 123 when the multifunction button 122 is pressed.

[0041] The battery 114 may be a disposable battery or a rechargeable battery in the form of an alkaline, nickel-cadmium, nickel-metal hydride battery, etc., such as any 3V-12 volts DC battery or other conventional battery, such as A, AA, AAA, etc., that supplies power to the automatic drawer-opening device 100. Throughout this specification, the term "battery" may be used interchangeably to refer to one or more wet or dry cells or batteries of cells in which chemical energy is converted into electricity and used as a source of DC power. References to recharging or replacing the battery 114 may refer to recharging or replacing individual cells, individual batteries of cells, or a package of multiple battery cells as is appropriate for any given battery technology that may be used. In addition, a rechargeable embodiment of the battery 114 may be recharged using a USB port 116, wherein the USB port 116 is a USB-A, USB-B, Micro-B, Micro-USB, Mini-USB, or USB-C port, etc.

[0042] As shown in FIGS. 2A-B, the housing component 102 is configured to house a processor 106 and a motor 104. Specifically, the housing component 102 is configured to house the processor 106 and motor 104 and any other circuitry as is known in the art for engaging the motor 104 to automatically open and close a kitchen drawer 110.

[0043] In one embodiment, the motor 105 is a DC motor but can be any suitable motor as is known in the art. In use, the motor 104 is installed within the spring-loaded wheel 108 and rotates the spring-loaded wheel 108 or other suitable device for contacting the drawer 110 to automatically open and/or close it.

[0044] Additionally, the spring-loaded wheel 108 is controlled by the motor 104 to rotate it, such that the wheel 108 can spin. Specifically, the wheel 108 is secured to a spring-loaded arm 200, which is tensioned to keep the arm 200 pushed against the cabinet 111, allowing the wheel 108 to contact the drawer 110. When activated by the motor 104, the wheel 108 spins to ride the cabinet enclosure 111 to open and close the drawer 110 hands-free. Then, once open, the motor 104 is engaged again to spin the spring-loaded wheel 108 in the opposite direction, to close the drawer 110 hands-free. Wheel 108 also includes an enclosure 109 which covers the wheel 108 and prevents things from getting sucked into the wheel 108.

[0045] In one embodiment, the spring-loaded wheel 108 has a rubber coating 202 for smooth operation and to protect the drawer 110 it comes into contact with. The rubber coating 202 also adds grip traction so the wheel 108 does not slip when rotating. In one embodiment, the spring-loaded wheel 108 can have adjustable tension, depending on the weight of the drawer 110 being opened and/or closed. Thus, the device 100 can accommodate different drawer weights. Specifically, the housing component 102 comprises a tension adjustment screw 204, which allows a user to turn the screw 204 right or left to provide for adjustable tension and user-specific customization.

[0046] As shown in FIG. 3, in one embodiment, multiple devices 100 and 101 can be linked together to operate in sync if the drawer 110 is large and heavy. One device 100 would be installed on the right side of the drawer 110 and a second device 102 on the left side of the drawer 110. With either the first 100 or the second 101 device, there may be a non-electrical spring-loaded wheel 108 which mounts opposite the electrical spring-loaded wheel 108, 104 to equalize pressing pressure of the spring-loaded wheels 108 and to ensure the devices 100, 101 do not bind and operate smoothly.

[0047] As shown in FIG. 4, the motor 104 is engaged via a hidden proximity sensor 400. Typically, the hidden proximity sensor 400 is positioned inside the drawer 110 face or on an outside of the drawer 110 to be activated by a user who comes in close proximity to the drawer 110 to be opened. The proximity sensor 400 can be any suitable sensor as is known in the art and can be wired or wireless. The proximity sensor 400 is in communication with the motor 104 and processor 106, such that once the sensor 400 is engaged, the sensor 400 communicates with the motor 104 to rotate the spring-loaded wheel 108.

[0048] As shown in FIGS. 5A-C, in one embodiment, the motor 104 is controlled by wireless controls. Specifically, the motor 104 can sync with smart features via Wi-Fi, Bluetooth, or virtual assistants. In this embodiment, the device 100 further includes a wireless communications module 502 and additional sensors which would allow the device 100 to pair with a mobile application 504 on a smart device 506. Once paired, a user could control the automatic drawer-opening device 100 via the mobile application 504. For example, the user can set open and close speeds in the mobile application 504. Further, the user can set notifications in the mobile application 504 when the drawer 110 opens. Thus, if a child were to open a drawer 110, the user would be alerted via the smart device 506. Additionally, the user can name drawers 110 in the mobile application 504, such as utensil drawer or snack drawer, etc. This would allow use of a virtual assistant to open a specific drawer 110, such as 'Alexa, open utensil drawer'. Typically, the smart device 506 may be a cellular telephone, a tablet, a computer, a remote control, or any other device that may have wireless communication capabilities and may be connected to the internet. The smart device 506 may perform any type of wireless communication, including, but not limited to, WIFI, BLUETOOTH, RFID, NFC, etc.

[0049] Further, the automatic drawer-opening device 100 can be installed at the point of manufacture or installation of the drawers 110. In another embodiment, the device 100 can be installed aftermarket, after the drawers 110 have been installed. Further, the device 100 is great for DIY installations and is easy-to-install, as needed. Thus, the automatic drawer-opening device 100 is great for users with grip issues, people with disabilities, or elderly users. Further, the device 100 eliminates the need for drawer pulls and displays a cleaner aesthetic, offering a clean, sleek appearance for drawers 110.

[0050] In yet another embodiment, the automatic drawer-opening device 100 comprises a plurality of indicia 500. The housing component 102 of the device 100 may include advertising, a trademark, or other letters, designs, or characters, printed, painted, stamped, or integrated into the housing component **102**, or any other indicia **500** as is known in the art. Specifically, any suitable indicia **500** as is known in the art can be included, such as but not limited to, patterns, logos, emblems, images, symbols, designs, letters, words, characters, animals, advertisements, brands, etc., that may or may not be drawer, hands-free, or brand related.

[0051] FIG. **6** illustrates a flowchart of the method of opening a drawer hands-free. The method includes the steps of at **600**, providing an automatic drawer-opening device comprising a housing component with a motor and spring-loaded wheel. The method also comprises at **602**, installing the housing component in a back of a drawer. Further, the method comprises at **604**, engaging the motor to rotate the spring-loaded wheel via a proximity sensor. The method also comprises at **606**, syncing the motor with smart features via Wi-Fi, Bluetooth, or virtual assistants. Finally, the method comprises at **608**, adjusting tension of the spring-loaded wheel to accommodate different drawer weights.

[0052] Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different users may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not structure or function. As used herein "automatic drawer-opening device", "drawer-opening device", "automatic device", and "device" are interchangeable and refer to the automatic drawer-opening device **100** of the present invention.

[0053] Notwithstanding the forgoing, the automatic drawer-opening device **100** of the present invention can be of any suitable size and configuration as is known in the art without affecting the overall concept of the invention, provided that it accomplishes the above stated objectives. One of ordinary skill in the art will appreciate that the automatic drawer-opening device **100** as shown in FIGS. **1-6** is for illustrative purposes only, and that many other sizes and shapes of the automatic drawer-opening device **100** are well within the scope of the present disclosure. Although the dimensions of the automatic drawer-opening device **100** are important design parameters for user convenience, the automatic drawer-opening device **100** may be of any size that ensures optimal performance during use and/or that suits the user's needs and/or preferences.

[0054] Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. While the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

[0055] What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are possible. Accordingly, the claimed subject matter is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term "includes" is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term "comprising" as "comprising" is interpreted when employed as a transitional word in a claim.

What is claimed is:

1. An automatic drawer-opening device that allows a drawer to be opened in a hands-free manner, the automatic drawer-opening device comprising:

a housing component;

a motor; and

a spring-loaded wheel; and

wherein the motor is housed within the spring-loaded wheel; and

further wherein the spring-loaded wheel contacts a drawer and rotates to automatically open the drawer hands-free.

2. The automatic drawer-opening device of claim **1**, wherein the housing component is configured in a rectangular shape to fit in a back of the drawer.

3. The automatic drawer-opening device of claim **2**, wherein the housing component comprises a plurality of mounting holes for allowing a user to insert a screw and mount the housing component in the back of the drawer.

4. The automatic drawer-opening device of claim **3**, wherein the motor is battery-operated.

5. The automatic drawer-opening device of claim **4**, wherein battery is replaceable.

6. The automatic drawer-opening device of claim **4**, wherein battery is rechargeable.

7. The automatic drawer-opening device of claim **6**, wherein rechargeable battery is recharged via a USB port positioned on the housing component.

8. The automatic drawer-opening device of claim **7**, wherein the housing component comprises a battery door, to allow a user access to the battery and a second port for plugging in a sensor.

9. The automatic drawer-opening device of claim **8**, wherein the housing component comprises a multicolor LED light and a multifunction button to control functions of the automatic drawer-opening device.

10. The automatic drawer-opening device of claim **9**, wherein the housing component is configured to house a processor and a motor in communication.

11. The automatic drawer-opening device of claim **10**, wherein the housing component comprises an LCD screen and a buzzer.

12. The automatic drawer-opening device of claim **11**, wherein the spring-loaded wheel comprises a wheel secured to a spring-loaded arm, such that the spring-loaded arm is under tension to keep the spring-loaded wheel pushed against the drawer.

13. The automatic drawer-opening device of claim **12**, wherein the spring-loaded wheel has a rubber coating.

14. The automatic drawer-opening device of claim **13**, wherein the spring-loaded arm has adjustable tension which is adjusted via a tension adjustment screw positioned on the housing component.

15. An automatic drawer-opening device that allows a drawer to be opened in a hands-free manner, the automatic drawer-opening device comprising:

a housing component configured in a rectangular shape to fit in a back of a drawer;

a DC motor which is battery-operated; and

a spring-loaded wheel comprising a wheel secured to a spring-loaded arm which allows the wheel to contact the drawer, which when activated by the motor, acts to rotate the wheel against the drawer to propel the drawer forward hands-free;

wherein the motor is housed within the spring-loaded wheel and a processor is housed within the housing component;

wherein the housing component comprises a plurality of mounting holes for allowing a user to insert a screw and mount the housing component in the back of the drawer;

wherein the housing component comprises a battery door, to allow a user access to a battery;

wherein the housing component comprises a multicolor LED light and a multifunction button to control functions of the automatic drawer-opening device;

wherein the spring-loaded wheel has a rubber coating;

wherein the motor engages the spring-loaded wheel to rotate via a hidden proximity sensor, to automatically open the drawer hands-free; and

further wherein the spring-loaded arm has adjustable tension which is adjusted via a tension adjustment screw positioned on the housing component.

16. The automatic drawer-opening device of claim **15**, wherein a second automatic drawer-opening device is secured on a left side of the drawer and the automatic drawer-opening device is secured on a right side of the drawer to sync and operate together if the drawer is large and heavy.

17. The automatic drawer-opening device of claim **15**, wherein the motor can sync with smart features via Wi-Fi, Bluetooth, or virtual assistants.

18. The automatic drawer-opening device of claim **17** further comprising a wireless communications module which would allow the automatic drawer-opening device to pair with a mobile application on a smart device, and once paired, a user could control the automatic drawer-opening device via the mobile application.

19. The automatic drawer-opening device of claim **15** further comprising a plurality of indicia.

20. A method of opening a drawer hands-free, the method comprising the following steps:

providing an automatic drawer-opening device comprising a housing component with a motor and spring-loaded wheel;

installing the housing component in a back of a drawer;

engaging the motor to rotate the spring-loaded wheel via a proximity sensor;

syncing the motor with smart features via Wi-Fi, Bluetooth, or virtual assistants; and adjusting tension of the spring-loaded wheel to accommodate different drawer weights.

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